

Enhancing Operational Efficiency in a Multispecialty Hospital

HealthFirst Care — Business Analysis Final Report

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Executive Summary

- Business Problem: HealthFirst Care faces a 73% appointment non-completion rate, 41.4-minute average wait times, and 43% resource unavailability across six departments, with patient satisfaction at 6.31/10.
- Root Cause: Manual, phone-based scheduling with no conflict detection drives double-bookings and no-shows.
- Data Evidence: 52 no-shows out of 216 appointments; only 57% of resources operational at any time.
- Business Impact: Resource unavailability is discovered only at the point of need, compounding operational delays.
- Key Insight: Scheduling failure and resource invisibility are the core bottlenecks — not staff capability.
- Recommended Solution: Deploy automated scheduling with conflict detection, SMS/email reminders, a real-time resource dashboard, and a unified HIS.

Introduction

- Business Context: HealthFirst Care is a multi-specialty hospital facing patient complaints and operational challenges.
- Opportunity: Transform manual hospital operations into an automated, integrated, data-driven model that improves patient experience.
- Data Sources: Analyze 646 records across appointment, feedback, and resource datasets to quantify operational failures.
- Analysis Approach: Apply Business Analysis frameworks — BRD, RTM, stakeholder analysis, process modeling, and risk management.
- Key Question #1: Will automation and conflict detection lift completion rates from 27% to 60%+?
- Key Question #2: Can real-time tracking cut resource unavailability from 43% to below 20%?

Business Objectives

Areas of improvement (6 measurable objectives):

- Reduce average patient wait times by 20% (41.4 → 33 minutes) within 6 months.
- Increase appointment completion rate from 27% to 60%+ within 6 months.
- Reduce resource unavailability from 43% to below 20% via real-time tracking.
- Eliminate scheduling double-bookings through automated conflict detection.
- Improve patient satisfaction from 6.31/10 to 8.0/10 within 12 months.
- Achieve full system integration of scheduling, records, billing, and communication.

BRD — Problem Statement

Problem Statement

HealthFirst Care suffers long wait times (41.4 min average), a 73% appointment non-completion rate, and approximately 43% resource unavailability. These failures are caused by manual scheduling, fragmented data silos, and poor inter-departmental communication across six departments: Cardiology, Orthopedics, Neurology, Oncology, Pediatrics, and General Medicine.

Quantified impact

- Only 58 of 216 appointments completed (27%); 52 no-shows, 50 rescheduled, 47 cancelled.
- Patient satisfaction averages 6.31/10; top complaints are delayed response and long wait times.
- Only 57% of 213 tracked resources are operational; Oncology and Pediatrics are worst affected.

BRD — Key Requirements

Key Requirements to improve operational efficiency

- Online self-service booking with automated double-booking conflict detection (FR-01, FR-02).
- Automated SMS/email reminders 24h and 2h before appointments (FR-03).
- Real-time resource dashboard with maintenance auto-reassignment (FR-04, FR-05).
- Inter-departmental communication portal for results, referrals, and transfers (FR-06, FR-07).
- Unified patient records / Hospital Information System across all departments (FR-08).
- Analytics reporting and billing-scheduling integration (FR-09, FR-10).

BRD — Constraints & Acceptance Criteria

Constraints

- Budget: upgrades must fit approved capital expenditure; phased rollout may be required.
- Timeline: project must complete within 12 months.
- Legacy systems and frequent network downtimes require extra development and stabilization.
- Staff resistance may slow adoption of new workflows.

Acceptance Criteria

- Wait time ≤ 33 min and completion rate $\geq 60\%$ within 6 months.
- Resource unavailability $< 20\%$; zero double-bookings in any 30-day period.
- Patient satisfaction $\geq 8.0/10$ within 12 months.
- 100% department portal adoption within 3 months of go-live.

Requirements Traceability Matrix (RTM)

Req ID	Requirement Description	Priority (MoSCoW)	Stakeholders	Project Objective	Related Data File	Status
FR-01	Online booking, reschedule, cancel without staff	Must	Patients	Reduce wait times (BO-01)	appointment_data.csv	Approved
FR-02	Auto-detect & prevent double-booking conflicts	Must	Admin Staff	Eliminate double bookings (BO-04)	appointment_data.csv	Approved
FR-03	Automated SMS/email reminders (24h & 2h)	Must	Patients	Increase completion to 60% (BO-02)	appointment_data.csv	Approved
FR-04	Real-time resource availability dashboard	Must	IT, Nurses	Reduce unavailability <20% (BO-03)	resource_data.csv	In Progress
FR-05	Flag maintenance & auto-reassign resources	Must	Doctors	Reduce unavailability <20% (BO-03)	resource_data.csv	In Progress
FR-06	Inter-departmental messaging portal	Must	Nurses, Admin	Full integration (BO-06)	—	Approved
FR-08	Unified patient records across departments	Must	Admin, IT	Full integration (BO-06)	—	Approved
FR-09	Reports on wait times, utilization, outcomes	Should	Management	Satisfaction 8.0/10 (BO-05)	feedback_data.csv	In Progress
NFR-03	Encrypt data; patch legacy vulnerabilities	Must	IT, Compliance	Full integration (BO-06)	—	Not Started
NFR-04	Accessible UI for low digital literacy	Should	Patients	Satisfaction 8.0/10 (BO-05)	feedback_data.csv	Not Started

Stakeholder Analysis — Stakeholders

Stakeholders

- Patients (Sarah Ayvazyan, Lak Ayer) — primary beneficiaries of care improvements.
- Doctors & Nurses (Dr. Khan, Dr. Lee, S. Murmu, J. Gomes) — clinical staff impacted by scheduling and resources.
- Administrative Staff (Maria Carter, Ivan Walker) — scheduling, billing, and records.
- IT Team (Rajesh Singh, Laura Simkow) — system integration and infrastructure.
- Hospital Management — project sponsor and decision-maker.

Stakeholder Analysis — Power / Interest Matrix

Stakeholders' influence mapped on the Power–Interest grid:

Influence \ Interest	Low Interest	High Interest
High Influence	Keep Satisfied: IT Team, Hospital Management	Key Players: Doctors, Nurses, Admin Staff
Low Influence	— (monitor)	Keep Informed: Patients

- Key Players (High power, High interest): Doctors, Nurses, Admin Staff.
- Keep Satisfied (High power, Low interest): IT Team, Hospital Management.
- Keep Informed (Low power, High interest): Patients.

Stakeholder Analysis — Engagement Strategies

Stakeholder Engagement Strategies

- Key Players — actively involve in decision-making; weekly meetings, dashboard access, requirement validation.
- Keep Satisfied — bi-weekly updates, executive summaries, technical sign-off, escalation alerts.
- Keep Informed — monthly newsletters, SMS/email notifications, satisfaction surveys.
- Engagement intensity matches each group's power and interest to build ownership and adoption.

Stakeholder Analysis — Communication Plan

Stakeholder Communication Strategies

- Doctors & Nurses: weekly meetings plus communication portal (owner: Project BA).
- Administrative Staff: weekly operational meetings, system training, scheduling-system access.
- IT & Hospital Management: bi-weekly email, KPI dashboards, escalation under a 48-hour protocol.
- Patients: monthly SMS/email notifications and satisfaction surveys (owner: Admin Staff).

Scope Management — In Scope / Out of Scope

In-Scope Activities

- Automated scheduling with conflict detection and SMS/email reminders.
- Real-time resource dashboard with maintenance auto-reassignment.
- Inter-departmental portal, unified HIS, self-service check-in kiosks.
- Analytics module, billing sync, legacy security patching, UAT, staff training.

Out-of-Scope Activities

- Clinical decision-support or diagnostic tools; new physical equipment procurement.
- Changes to clinical protocols; third-party external system integration.
- Patient mobile app (future phase).
- Hiring, recruitment, or HR/compensation changes.

Scope Management — Assumptions

Assumptions

- Management releases budget for upgrades and cloud infrastructure in month 1.
- All six departments participate in user acceptance testing (UAT) before go-live.
- Existing patient data migrates to the new HIS without loss.
- IT team is available throughout the implementation phase.
- Network is stabilized before deployment; staff complete training pre-launch.

Scope Management — Constraints

Constraints

- Fixed capital budget; phased rollout may be required.
- 12-month timeline aligned to the annual operational review cycle.
- Outdated legacy systems requiring additional development time.
- Frequent network downtime requiring infrastructure stabilization.
- Regulatory compliance required before go-live; change-management challenges from staff adoption.

Scope Management — WBS Phases

Phases in the Work Breakdown Structure (WBS):

WBS ID	Task Name	Task Description	Milestone
1.1	Project Initiation & Planning	Charter, stakeholder plan, BRD/RTM finalization, schedule	M1: Approved (Month 1)
1.2	Requirements Gathering & Analysis	Interviews, data analysis, As-Is/To-Be design, validation	M2: Baselined (Month 2)
1.3	System Design & Development	Build scheduling, dashboard, portal, HIS, kiosks, cloud, security	M3: Dev Complete (Month 7)
1.4	Testing, UAT & Deployment	Unit/integration/NFR testing, UAT, training, migration, go-live	M4: Go-Live (Month 9)
1.5	Monitoring, Evaluation & Closure	KPI tracking, satisfaction surveys, scope validation, closure	M5: Closed (Month 12)

Scope Management — Change Management

Scope Change Management

All changes follow a formal Scope Change Request (SCR) process — verbal or email requests are not valid until formally submitted.

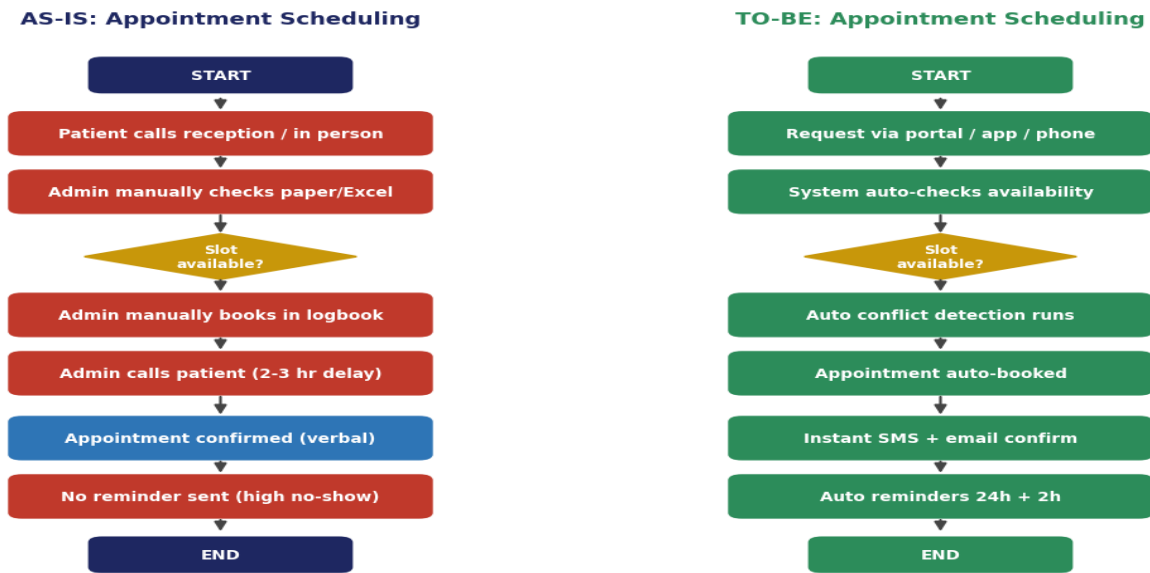
- Identify change → submit SCR form → impact assessment by BA and IT Manager within 5 days.
- Change Control Board (CCB) reviews; majority vote to approve, reject, or defer.
- Hospital Management holds veto on budget- or timeline-impacting changes.
- If approved: update BRD, RTM, WBS, and schedule; communicate to stakeholders; implement and monitor.
- Approval requires strategic alignment, budget and timeline feasibility, stakeholder support, compliance, and RTM traceability.

Process Mapping — As-Is / To-Be

Process	As-Is Model	To-Be Model
Appointment Scheduling	Manual phone booking, no conflict detection, 2–3 hr confirmation delay, no reminders (52 no-shows).	Multi-channel online booking, real-time conflict detection, instant SMS/email confirmation and reminders.
Patient Check-In	20+ min queues, manual paper forms, slow record verification, no department notification.	Self-service kiosk / online check-in, auto-verification <10 sec, instant department alert, <5 min check-in.
Interdepartmental Communication	Untracked phone/email requests, no SLA, no status updates, no audit trail.	Structured dashboard, auto-prioritized SLA, instant IT alerts, full audit trail and analytics.

Advanced Process Mapping — BPMN Workflow

Detailed Appointment Scheduling Workflow (BPMN): As-Is vs. To-Be



As-Is BPMN (left): patient calls reception; admin manually checks paper/Excel schedule; no conflict detection; manual logbook booking; admin calls to confirm with a 2-3 hr delay; verbal-only confirmation; no reminder sent; high no-show risk.

To-Be BPMN (right): patient requests via online portal/app/phone; system auto-checks real-time availability; automated conflict detection prevents double-booking; appointment auto-booked; instant SMS and email confirmation; automated 24h and 2h reminders; doctor/department auto-notified.

The As-Is BPMN highlights manual scheduling with no conflict detection, leading to high no-show risk. The To-Be BPMN illustrates an automated, real-time booking process with instant conflict detection and reminders.

Advanced Process Mapping — Swimlane Diagram

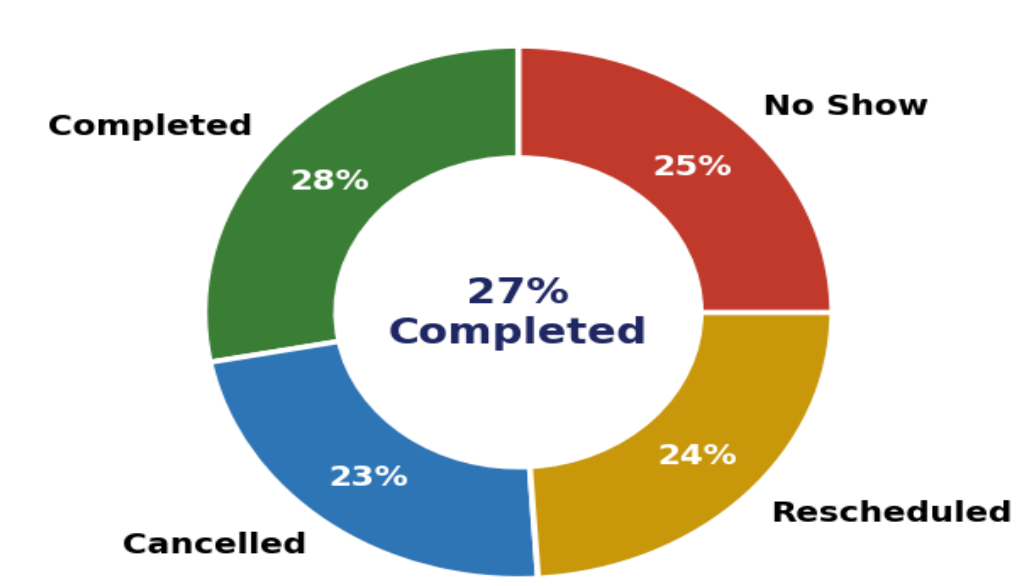
Stakeholder responsibility using the Swimlane diagram:

Swimlane (Stakeholder)	Task / Activity	Description
Patient	Request appointment	Books via portal/app/phone; receives confirmation and reminders.
Admin Staff	Validate & schedule	System auto-checks availability; admin handles exceptions only.
Scheduling System	Conflict detection	Runs automated conflict check; prevents double-booking.
Doctor / Department	Receive notification	Auto-notified of upcoming appointment; reviews patient record.
Nurse	Resource preparation	Checks real-time dashboard; prepares room and equipment.
IT Team	System monitoring	Maintains uptime, security patches, and integration health.
Billing	Auto-sync	Scheduling data syncs to billing; reconciliation automated.
Management	KPI oversight	Monitors dashboards; approves scope and budget changes.

Data Analysis — Pivot Table & Trends

Trends using a Pivot Table — Appointment Status (216 records):

Appointment Status Breakdown (216 records)



Pivot Table — Appointment Status

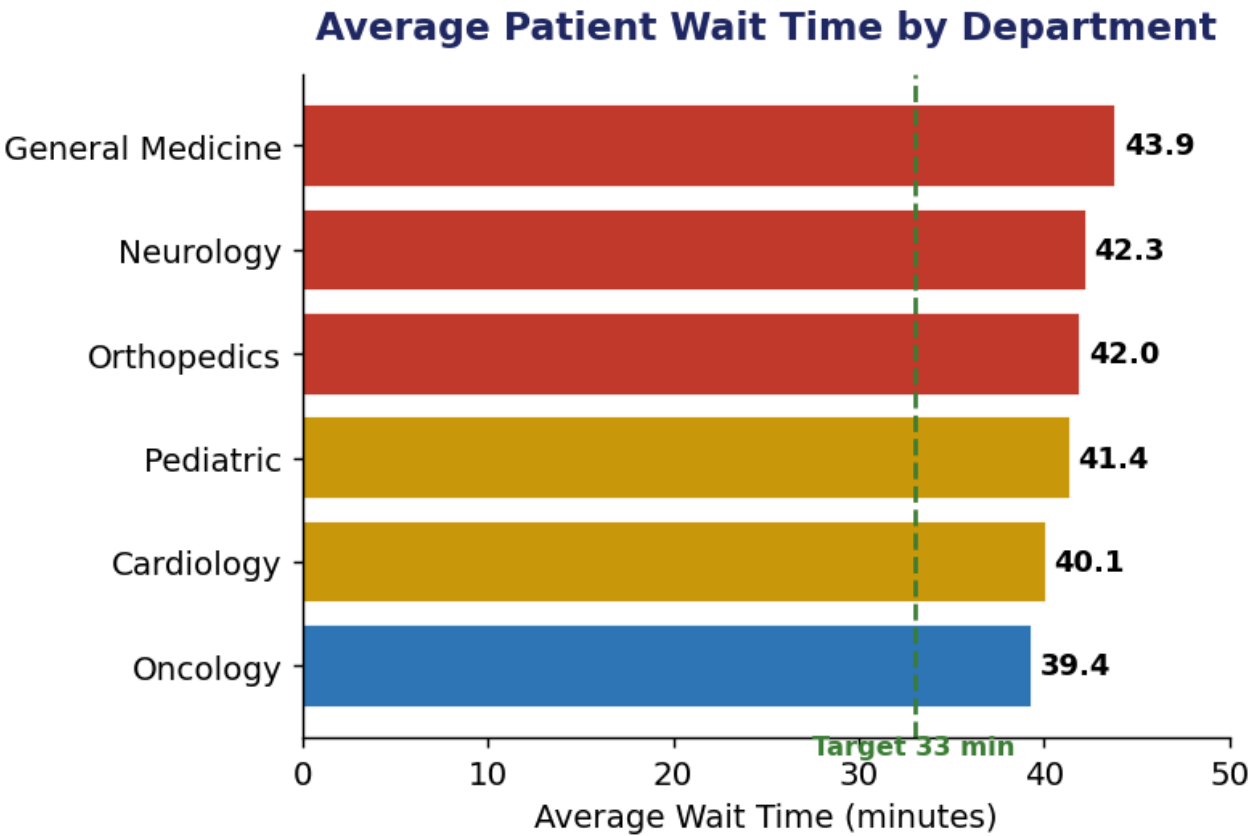
This pivot table summarizes appointment status across 216 records: Completed 58 (27%), No Show 52 (24%), Rescheduled 50 (23%), and Cancelled 47 (22%). The trend shows a 73% non-completion rate. Volume by department is highest in Pediatrics (39), Cardiology (38), and Neurology (38). Resources are only 57% operational, with 65 unavailable (31%) and 26 under maintenance (12%). Average satisfaction is 6.31/10 and average wait time is 41.4 minutes.

Data Analysis — Key Insights

Key Insights from the data analysis

- Scheduling failure is the core problem — the 73% non-completion rate stems directly from manual booking, no conflict detection, and no reminders.
- Resource invisibility compounds delays — unavailability is discovered only at the point of need, with no real-time tracking.
- General Medicine and Neurology drive the longest waits (43.9 and 42.3 min) and the lowest satisfaction.
- Positive feedback shows staff capability exists — the bottleneck is systems, not people.
- Automation, real-time dashboards, and integration directly target every quantified pain point.

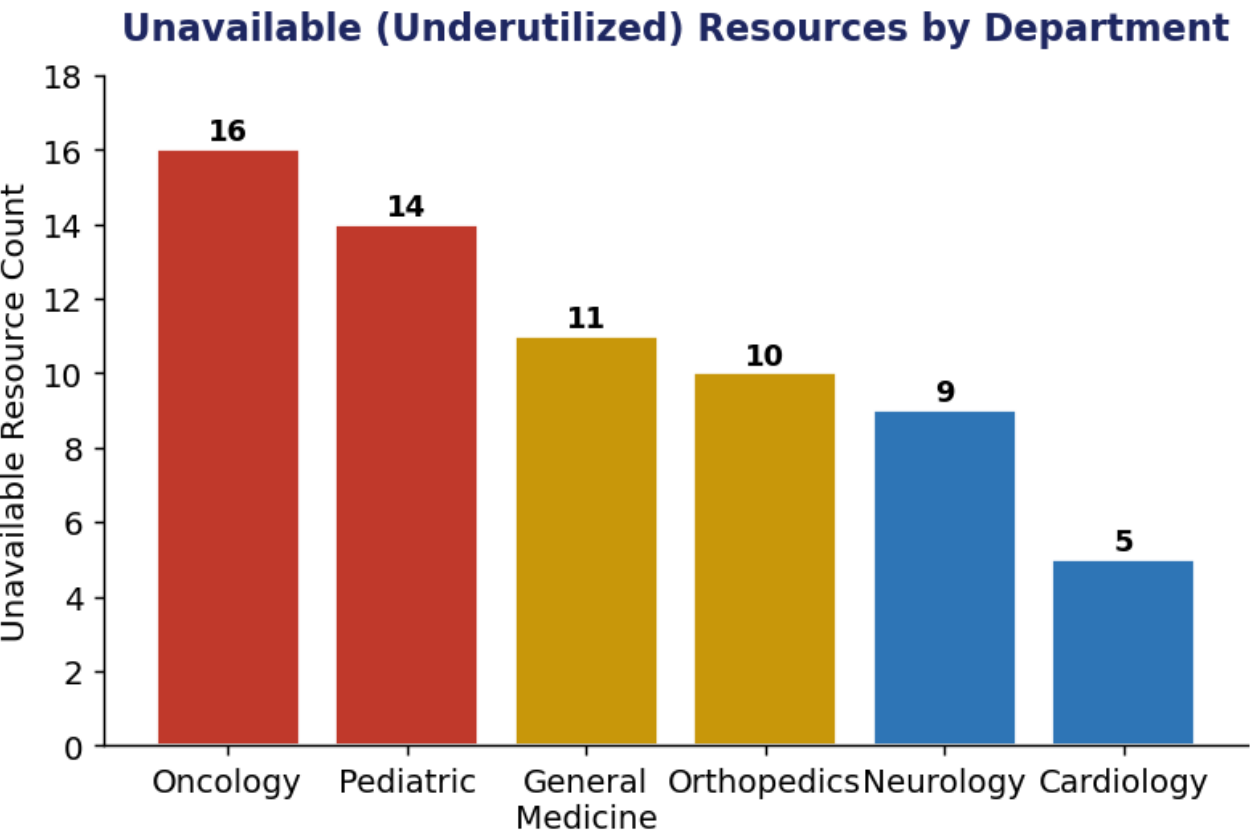
Data Visualization — Patient Wait Time (Bar Chart)



Insight

This horizontal bar chart shows average patient wait time by department. General Medicine (43.9 min), Neurology (42.3 min), and Orthopedics (42.0 min) exceed the 33-minute target most severely, while Oncology (39.4 min) and Cardiology (40.1 min) are closest. Every department currently sits above the target line, confirming wait time is a system-wide problem requiring automated scheduling.

Data Visualization — Resource Utilization (Bar Chart)

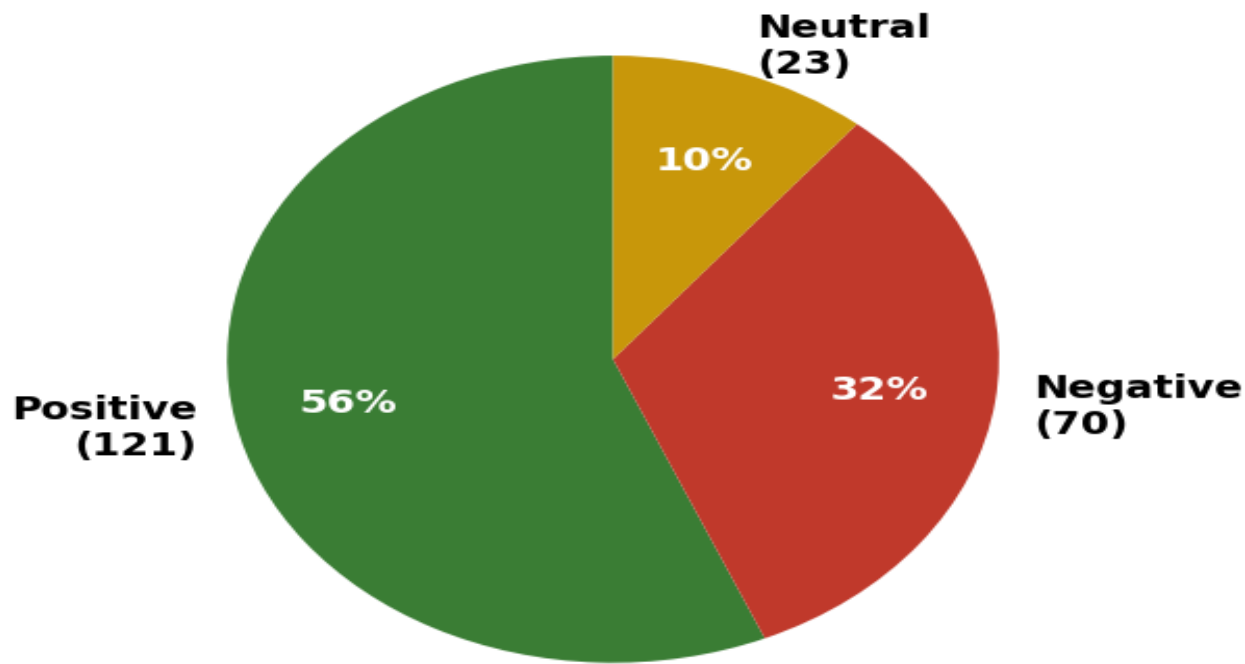


Insight

This bar chart highlights overused and underutilized resources by department, measured as unavailable resource count. Oncology (16) and Pediatrics (14) carry the highest unavailability, followed by General Medicine (11) and Orthopedics (10). Cardiology (5) is least affected. The pattern shows resource shortages concentrate in high-volume departments, justifying a real-time resource dashboard.

Data Visualization — Patient Feedback (Pie Chart)

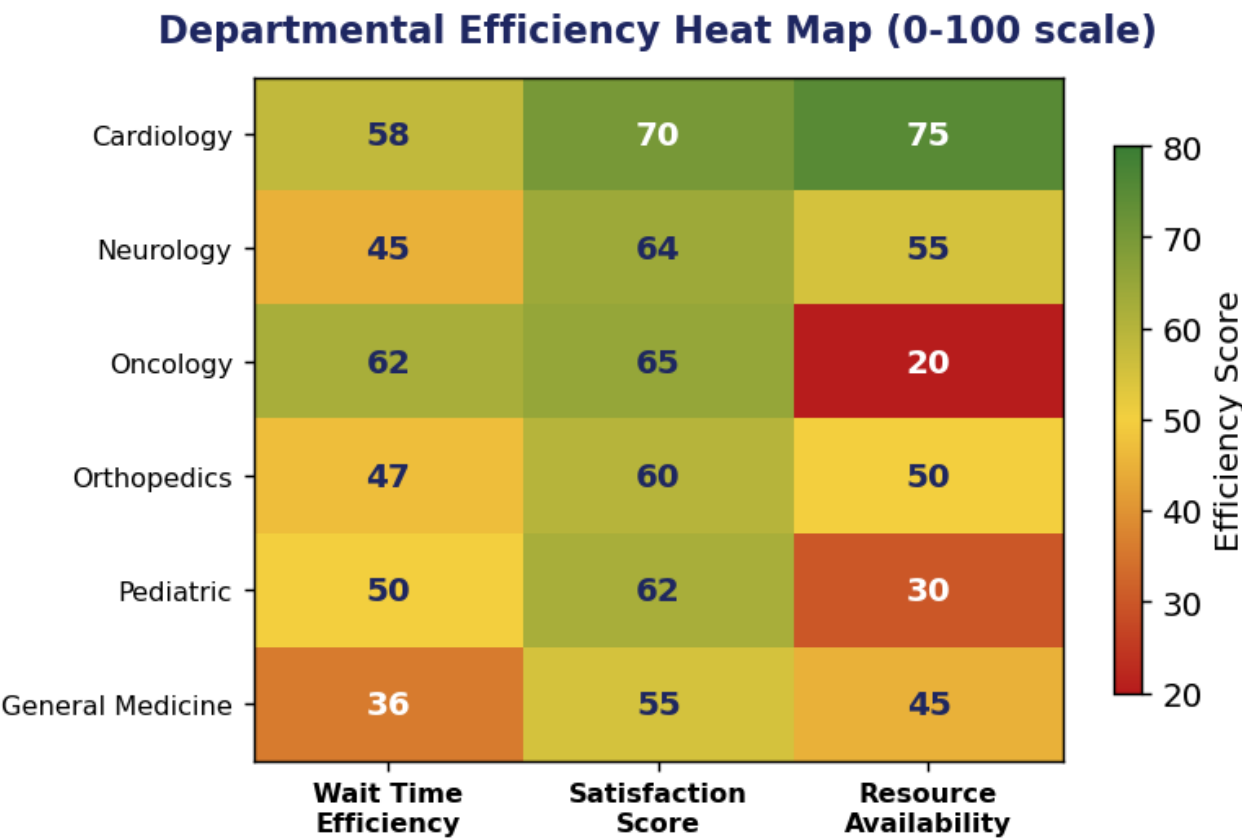
Patient Feedback Sentiment Distribution



Insight

This pie chart visualizes patient feedback sentiment: Positive 57% (121), Negative 33% (70), and Neutral 11% (23). While a majority is positive, the one-third negative share aligns with delayed-response and long-wait complaints, and confirms that targeted scheduling and communication fixes can raise the 6.31/10 satisfaction score toward the 8.0/10 goal.

Data Visualization — Departmental Efficiency (Heat Map)



Insight

This heat map shows departmental efficiency across wait time, satisfaction, and resource availability on a 0–100 scale. Oncology resource availability (20) and Pediatrics (30) are critical red zones, while Cardiology scores strongest overall (75 availability). General Medicine has the lowest wait-time efficiency (36). The map pinpoints exactly where interventions deliver the most impact.

Risk Assessment — Risk Register

Risk ID	Risk Description	Category	Likelihood	Impact	Severity	Mitigation Strategy
R-01	Legacy system integration delays development & causes migration errors	Technical	High	High	High	Early legacy audit; WBS 1.3 buffer; phased migration with validation (NFR-03)
R-02	Cloud systems fail due to frequent network downtime	Technical	High	High	High	Stabilize network as prerequisite; redundant connectivity; offline kiosk fallback
R-03	Unpatched legacy vulnerabilities cause a data breach	Technical	Medium	High	High	Patch before go-live; encrypt at rest/transit; penetration test (NFR-03, NFR-06)
R-04	Staff resistance slows adoption & utilization	Stakeholder	High	Medium	High	Change management; early Key-Player involvement; role-based training; champions
R-05	Project exceeds approved budget, forcing scope cuts	Operational	Medium	High	High	Phased rollout; bi-weekly tracking; CCB approval; contingency reserve
R-06	12-month timeline slips from dependencies & complexity	Operational	Medium	Medium	Medium	Milestone gates M1–M5; critical-path monitoring; 48-hr escalation
R-07	Elderly/low-literacy patients struggle with kiosks	Stakeholder	Medium	Medium	Medium	Accessible UI (NFR-04); retain phone option; usability testing (TC-14)
R-08	Patient data lost/corrupted during HIS migration	Technical	Low	High	Medium	Full backup; post-migration reconciliation; staged cut-over with rollback
R-09	Inaccurate source data weakens analytics	Operational	Medium	Low	Low	Data cleansing pre-go-live; fix 9 missing slots; entry validation rules
R-10	Uneven portal adoption leaves coordination gaps	Stakeholder	Low	Medium	Low	Mandate 100% use in 3 months; track in KPIs; weekly reinforcement

Risk Assessment — Risk Matrix

Risks categorized based on the Risk Assessment Matrix:

Likelihood \ Impact	Low Impact	Medium Impact	High Impact
High Likelihood		High — R-04	High — R-01, R-02
Medium Likelihood	Low — R-09	Medium — R-06, R-07	High — R-03, R-05
Low Likelihood	Low	Low — R-10	Medium — R-08

High-priority risks (R-01 to R-05) require immediate attention. Medium risks (R-06, R-07, R-08) are monitored; low risks (R-09, R-10) are reviewed regularly.

Risk Assessment — SWOT Analysis

Strengths

- Strong management sponsorship & funding
- 646-record reliable data foundation
- Clear traceable requirements (BRD + RTM)
- Engaged Key Players validating workflows

Weaknesses

- Manual phone-based workflows, no conflict detection
- Fragmented data silos across six departments
- Outdated legacy systems with vulnerabilities
- 57% operability, 73% non-completion rate

Opportunities

- Automation to cut wait times & no-shows
- Unified HIS integrating all systems
- Real-time dashboard raising availability >80%
- Kiosks & training raising satisfaction

Threats

- Data breach from unpatched vulnerabilities
- Network downtime disrupting cloud services
- Staff resistance limiting adoption
- Budget/timeline overruns forcing scope cuts

Risk Assessment — Key Insights

Key Insights from the Risk Assessment

- Five high-severity risks (R-01 to R-05) require immediate attention: legacy integration, network instability, data security, staff resistance, and budget pressure.
- Technical risks dominate the high-severity tier — infrastructure readiness is the single biggest threat to go-live.
- Every mitigation is traceable to a WBS work package and RTM requirement, ensuring accountability.
- Risks are reviewed at each milestone gate (M1–M5) and escalated through the formal CCB process.
- The SWOT confirms strong internal foundations offset by legacy-system weaknesses.

Risk Mitigation — Strategies

Strategies to mitigate the five high-severity risks:

Risk ID	Risk Description	Severity	Mitigation Strategy
R-01	Legacy integration delays & migration errors	High	Early legacy audit; WBS 1.3 buffer; phased migration with validation (NFR-03)
R-02	Cloud systems fail due to network downtime	High	Stabilize network as prerequisite; redundant connectivity; offline kiosk fallback
R-03	Unpatched vulnerabilities cause a data breach	High	Patch before go-live; encrypt at rest/transit; penetration test (NFR-03, NFR-06)
R-04	Staff resistance slows adoption	High	Change management; early Key-Player involvement; role-based training; champions
R-05	Project exceeds approved budget	High	Phased rollout; bi-weekly tracking; CCB approval; contingency reserve

Risk Mitigation — Contingency Plan

Factors included in the Contingency Plan:

Risk ID	Contingency Plan & Recovery Target
R-01	Roll back to staged backup; isolate affected module; run reconciliation; engage integration vendor; extend dev via CCB. Recovery: 5 business days.
R-02	Switch kiosks/scheduling to offline mode; queue locally for sync; activate redundant connectivity; notify IT. Recovery: 24 hours.
R-03	Isolate affected systems; notify IT & Compliance; root-cause analysis; emergency fixes; reset credentials; breach notification (NFR-06). Recovery: 48 hours to containment.
R-04	Introduce department champions; additional hands-on training; gather feedback on pain points; escalate blockers to Management. Recovery: within 4 weeks of go-live.
R-05	Freeze non-critical scope; re-sequence rollout to defer low-priority modules; draw contingency reserve; submit revised budget. Recovery: next bi-weekly cycle.

Risk Mitigation — Risk Prioritization Matrix

Risks prioritized based on the Visual Risk Matrix:

Priority	Risk ID & Description	Rationale	Action Urgency
High	R-01, R-02 (Legacy integration, Network failure)	Highest likelihood + impact; block go-live entirely	Immediate
High	R-03 (Data breach)	Severe compliance & reputational impact	Immediate
High	R-04 (Staff resistance)	High likelihood; undermines ROI of all systems	Immediate
High	R-05 (Budget overrun)	Forces scope cuts; threatens objectives	Immediate
Medium	R-06, R-07, R-08 (Timeline, Usability, Migration)	Manageable with monitoring & planning	Monitor & manage
Low	R-09, R-10 (Data quality, Portal adoption)	Limited impact; addressed via routine process	Regular review

Risk Mitigation — Key Insights

Key Insights from the Risk Mitigation Plan

- Mitigation reduces exposure before risks occur; contingency plans ensure rapid recovery if they materialize.
- All five high-priority risks have a defined owner, recovery target, and traceable mitigation action.
- Network stabilization and legacy patching are prerequisites — they gate the entire go-live.
- Recovery targets range from 24 hours (network) to 5 days (integration), keeping disruption bounded.
- Continuous governance: milestone-gate reviews, CCB change control, and bi-weekly KPI reporting to Management.

Key Findings

1. Scheduling is the primary failure point

Only 27% of appointments complete, with 52 no-shows driven by manual booking and zero reminders.

2. Resource availability is critically low

Just 57% operational, with Oncology and Pediatrics most affected and no real-time visibility.

3. Wait times and satisfaction are linked

41.4-min average waits correlate with the 6.31/10 score; General Medicine performs worst on both.

4. Communication gaps are systemic

Untracked phone/email requests across six departments leave no audit trail and slow resolution.

Key Recommendations

- Automate scheduling with real-time conflict detection and SMS/email reminders to lift completion to 60%+.
- Deploy a real-time resource dashboard with maintenance auto-reassignment to raise availability above 80%.
- Integrate a unified HIS and communication portal connecting scheduling, records, and billing across all departments.
- Add self-service kiosks and accessible UI to cut check-in from 20+ minutes to under 5, including support for elderly patients.

Implementation: a phased 12-month rollout with stabilized network and patched legacy systems as go-live prerequisites.

Conclusion

- Data across 646 records confirms three core failures: 73% appointment non-completion, 41.4-min waits, and 43% resource unavailability.
- The root cause is systemic — manual workflows, data silos, and outdated legacy systems — not staff capability.
- Automation, real-time tracking, and full integration directly address every quantified pain point and business objective.
- A structured risk plan with mitigation and contingency actions keeps the 12-month initiative resilient.
- Successful delivery is projected to raise satisfaction to 8.0/10 and create a coordinated, efficient hospital environment.

Appendix — Supporting Materials

- Source data: appointment_data.csv (216), feedback_data.csv (217), resource_data.csv (213) — 646 records total.
- Full deliverables: Business Requirements Document (BRD), Requirements Traceability Matrix (RTM), Stakeholder Analysis & Engagement Plan.
- Process Model (As-Is / To-Be), Scope Management Plan with WBS, and Risk Management & Mitigation Plans.
- Charts: appointment status, wait time by department, resource utilization, feedback sentiment, and departmental efficiency heat map.